Towards Criticality-based Value Prediction

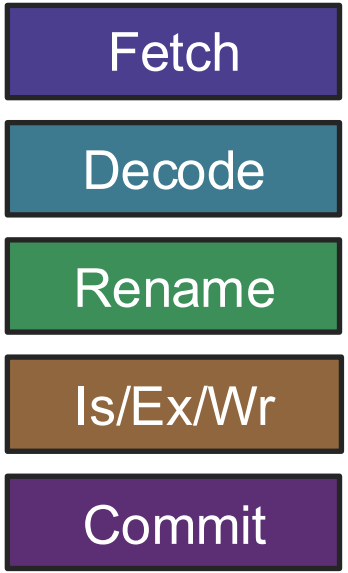
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https://dse.in.tum.de/



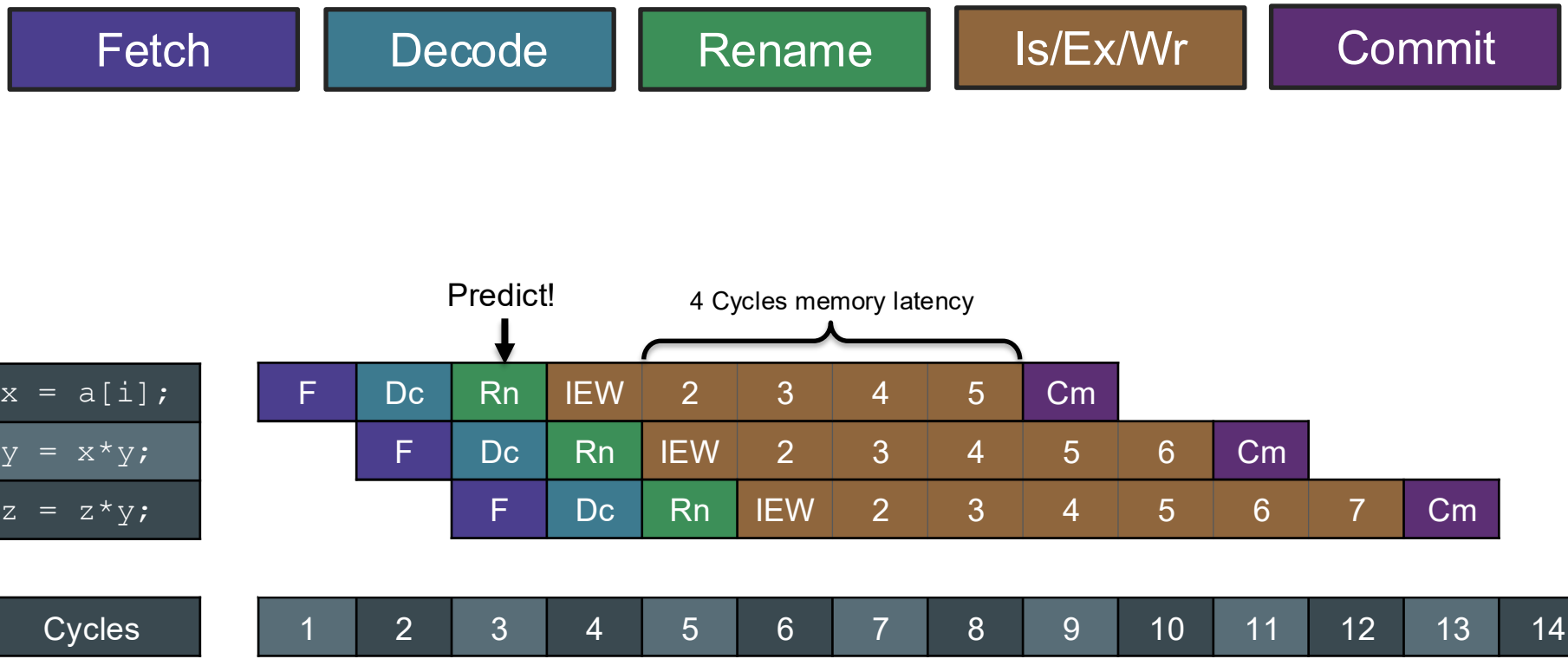
16.04.2026 – 17.08.2026

Value Predictor

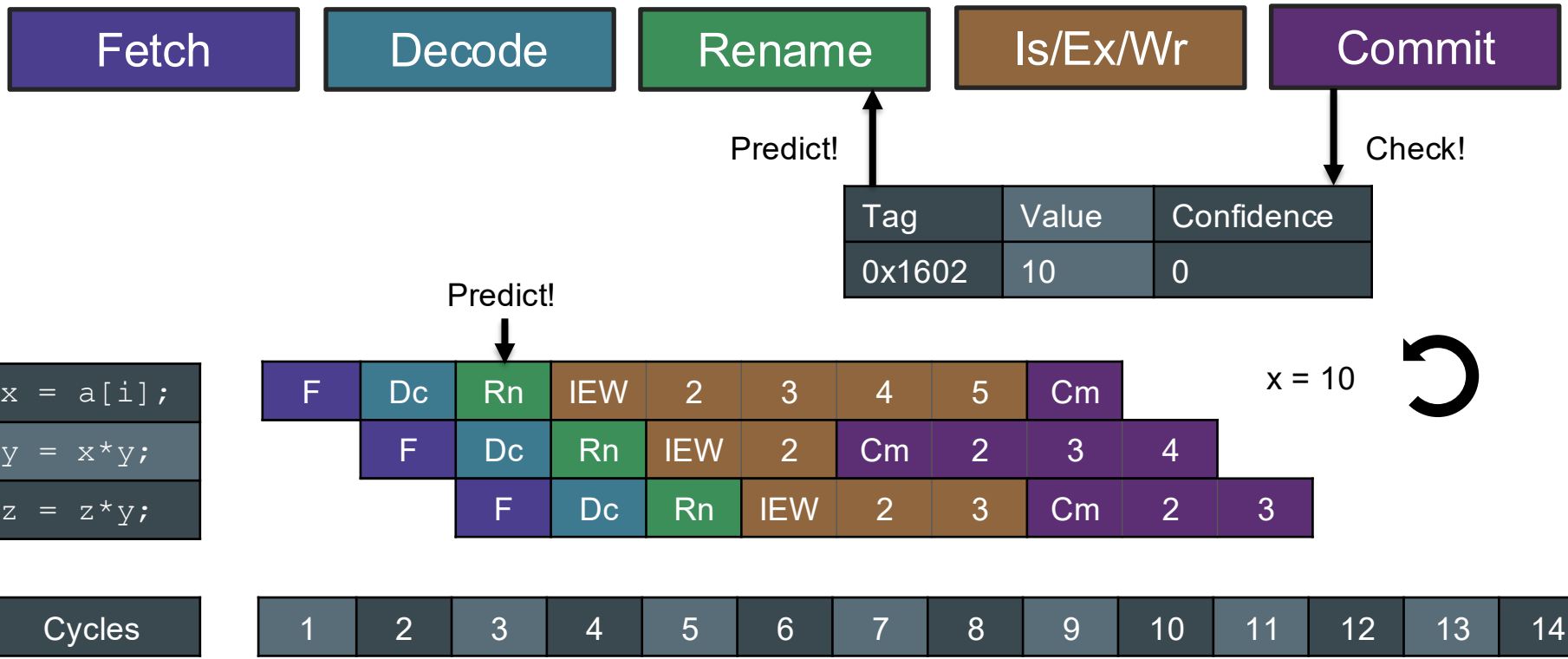
```
1  x = a[i];
2  y = x*y;
3  z = z*y;
```



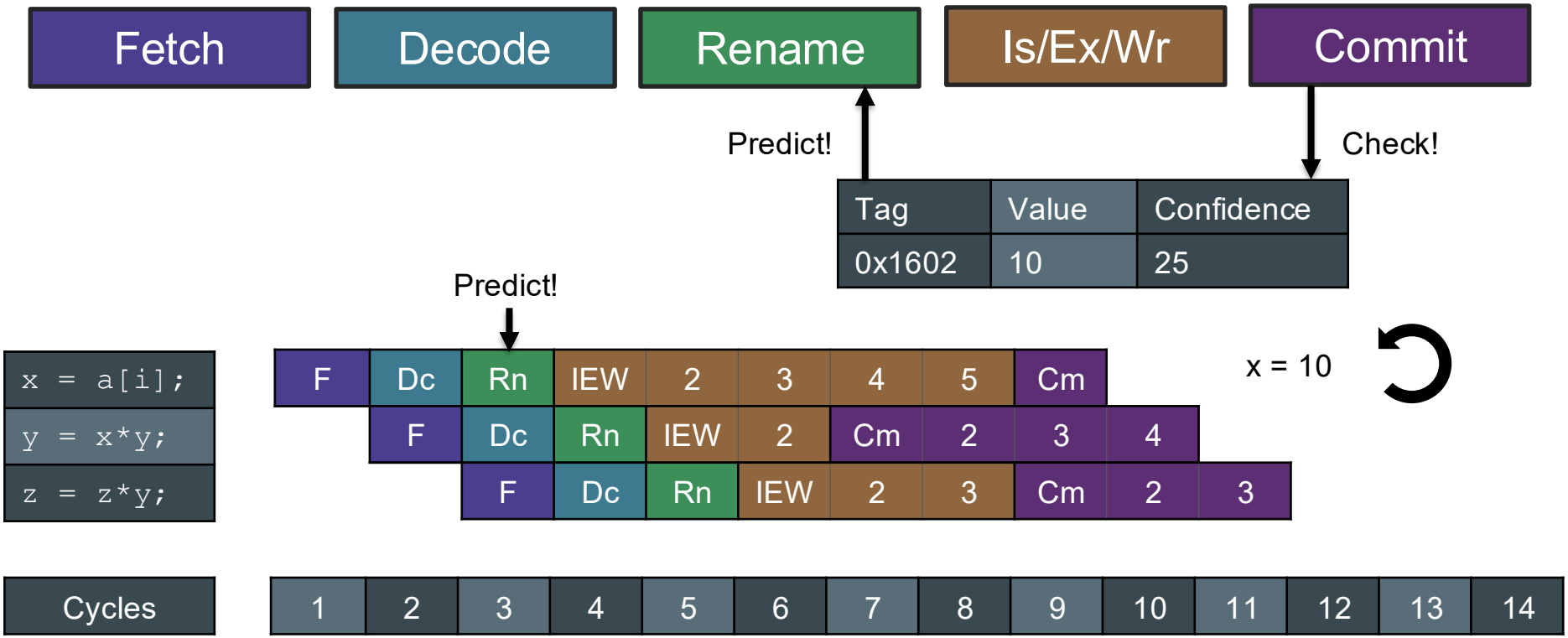
Value Predictor



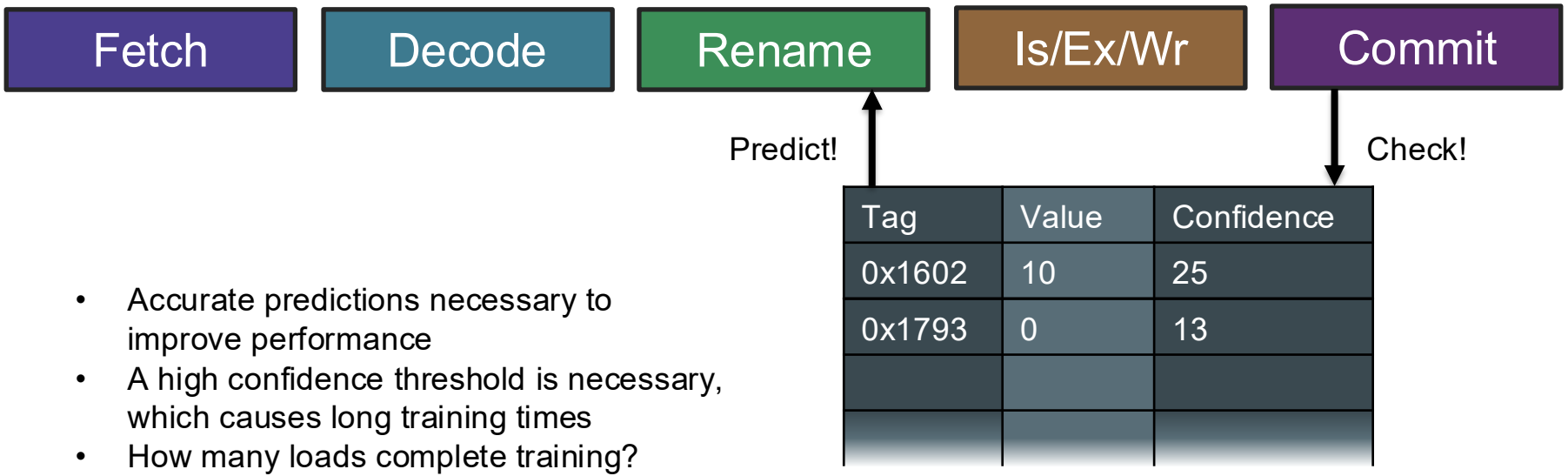
Value Predictor



Value Predictor

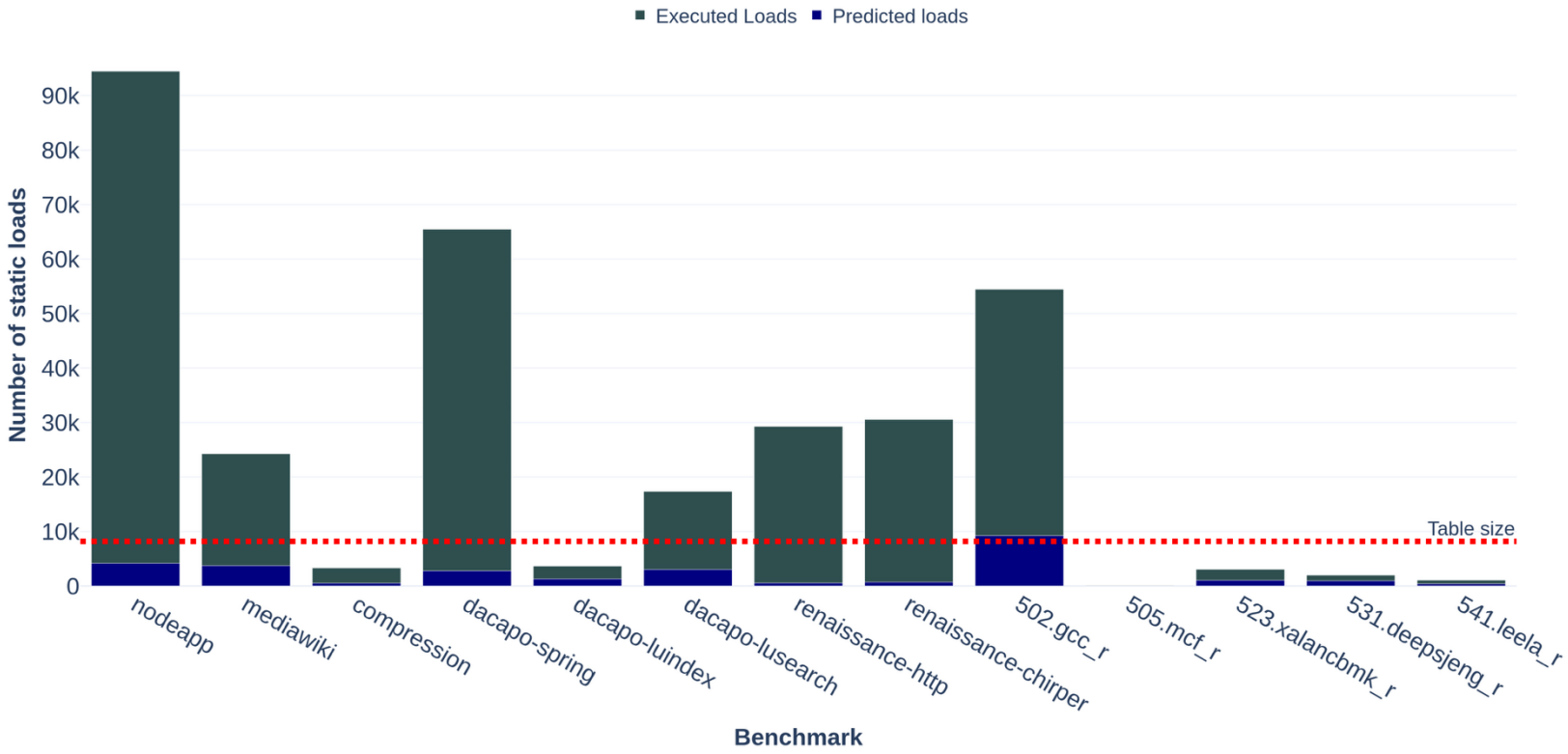


Value Predictor

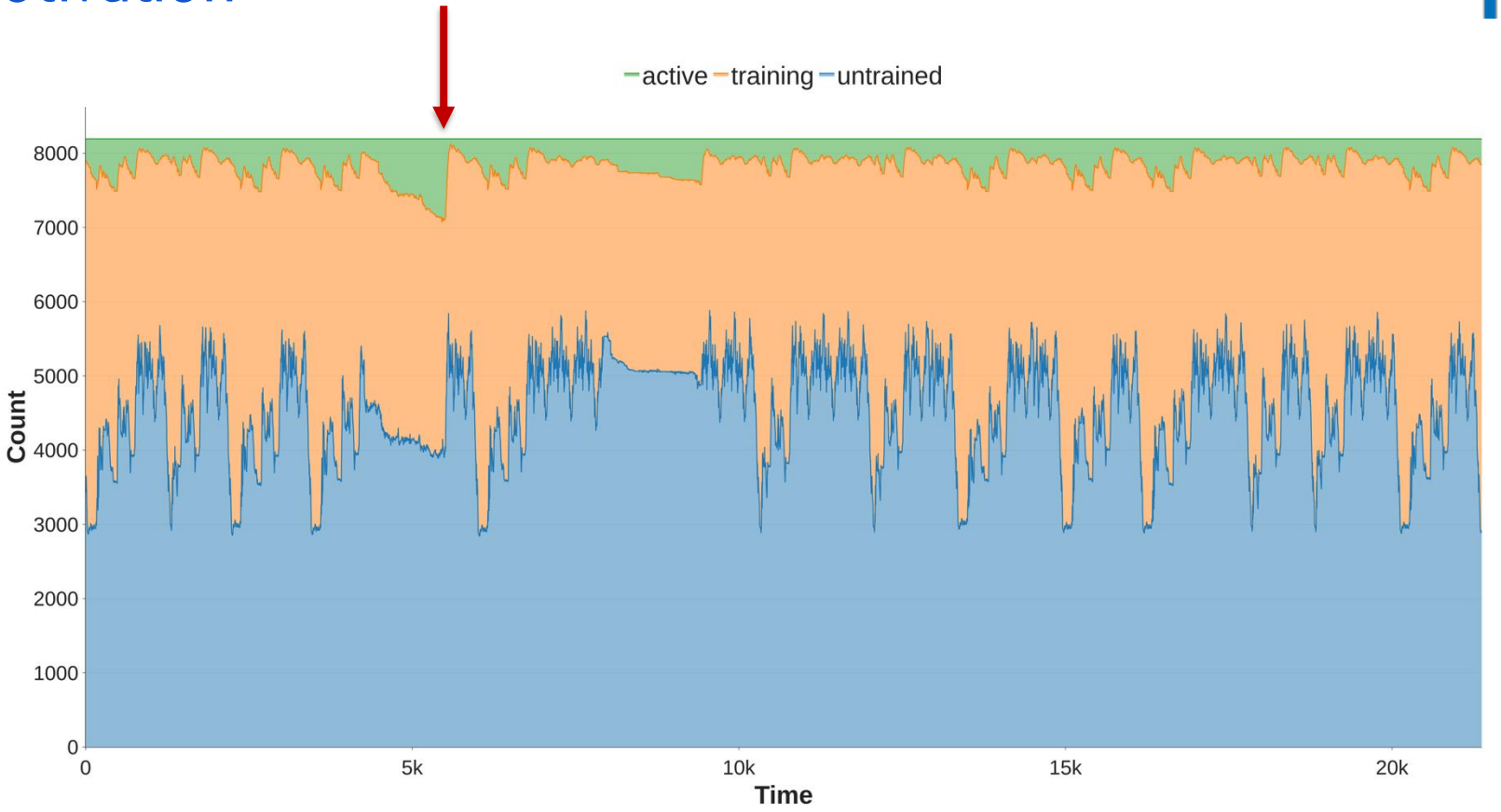


- Accurate predictions necessary to improve performance
- A high confidence threshold is necessary, which causes long training times
- How many loads complete training?

Motivation


Executed Loads
Predicted loads
Number of static loads
90k
80k
70k
60k
50k
40k
30k
20k
10k
0
Table size
nodeapp
mediawiki
compression
dacapo-spring
dacapo-luindex
dacapo-lusearch
renaissance-http
renaissance-chirper
502.gcc_r
505.mcf_r
523.xalancbmk_r
531.deepsjeng_r
541.leela_r
Benchmark

Motivation


active
training
untrained
Count
8000
7000
6000
5000
4000
3000
2000
1000
0
0
5k
10k
15k
20k
Time

Problem and Design

Problem

- A lot of the loads never complete training and therefore waste space
- If too many new loads occur, the valuable training process is lost

Design Idea

- Only want to train loads that are likely to complete training
- Need to filter out loads that are unpredictable

-> Put a filter in front of the prediction table that is based on predictability

Filter Implementation

- Implemented as a small prediction table
- Filter runs a short training phase and advances entries that reach the entry threshold

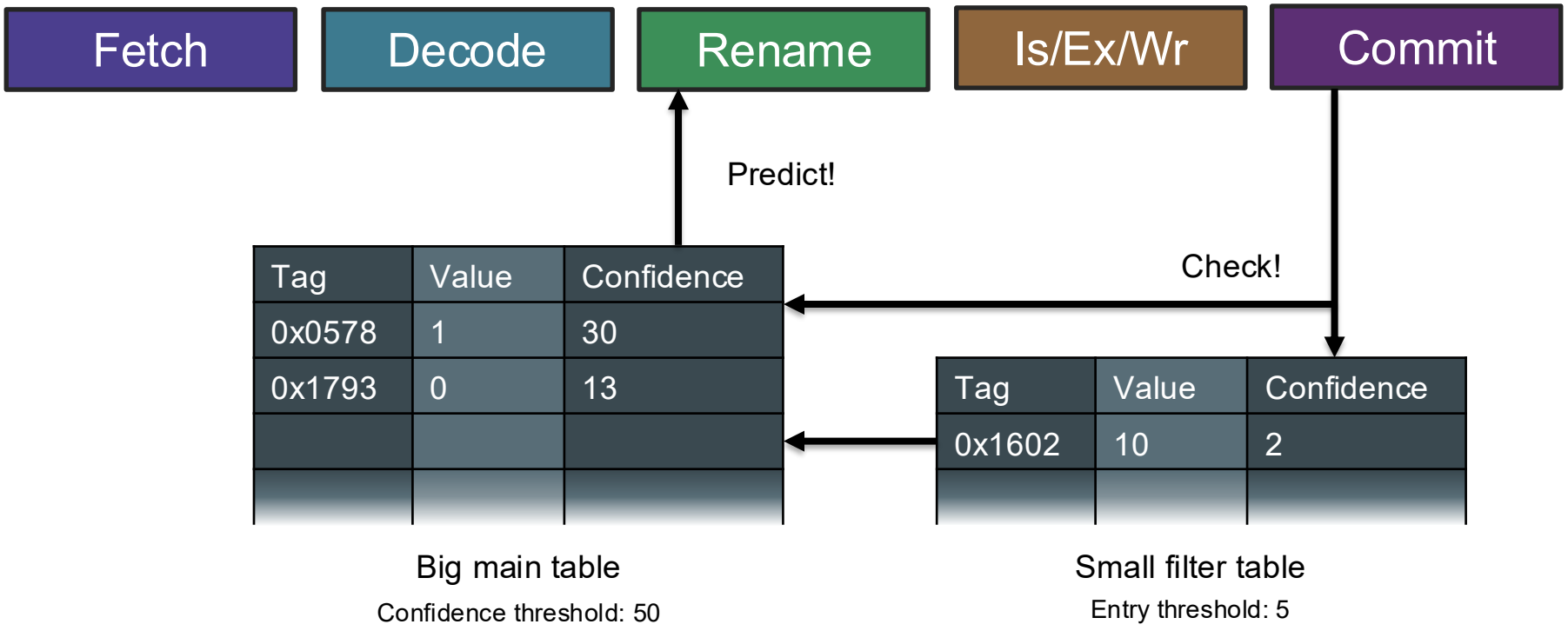
- Prefers loads that are predictable and therefore increase their confidence fast
- Stops many new loads from flooding the big prediction table

Tag	Value	Confidence
0x1602	10	2

Small filter table

Entry threshold: 5

2-level Value Predictor Implementation



Methodology

Benchmark
nodeapp
mediawiki
compression
dacapo-spring, dacapo-luindex, dacapo-lusearch
renaissance-http, renaissance-chirper
502.gcc_r, 505.mcf_r, 523.xalancbmk_r, 531.deepsjeng_r, 541.leela_r

Benchmarks

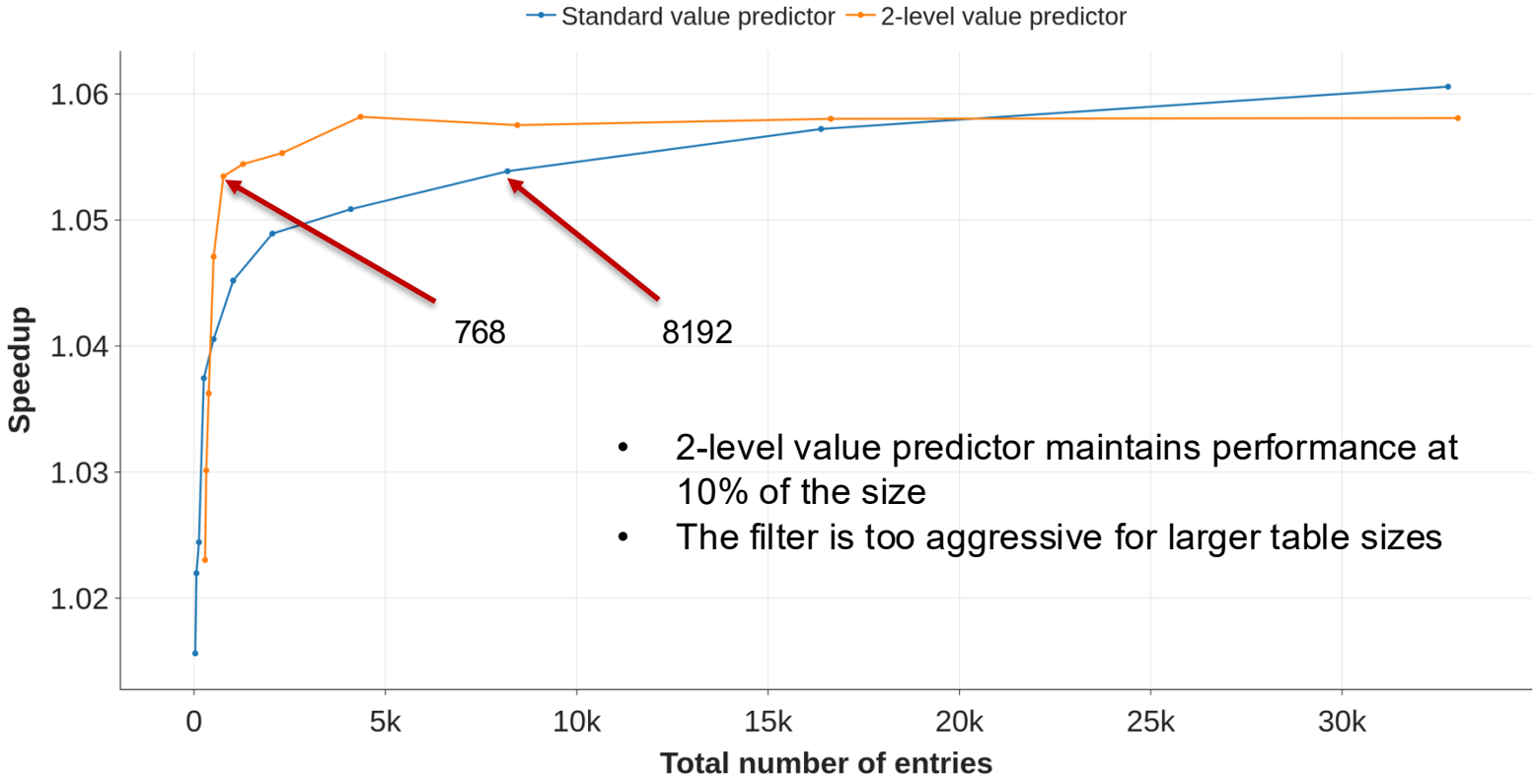
System	Full-system ARM
Core	3GHz, 8-wide OoO, 567-entry ROB, 189-entry LQ, 120-entry SQ
Cache Hierarchy	L1I: 64KiB 16-way, L1D: 48KiB 12-way, L2: 3MiB 12-way, L3 8MiB 16-way
BTB	16K entry, 8-way associative
Branch Predictor	64KB TAGE-SC-L
Prefetcher	Data: Stride, Instructions: Fetchdirected + Tagged

gem5 CPU configurations

Value predictor configurations

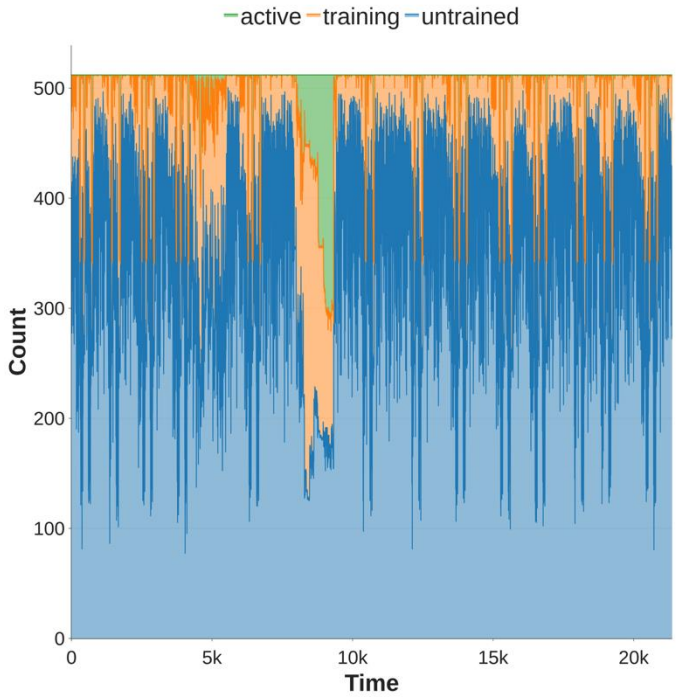
- Standard value predictor: Stride predictor with confidence threshold 50
- 2-level value predictor: 256-entry entry threshold 4 filter with confidence threshold 50 stride predictor
- Baseline: No predictor

Evaluation

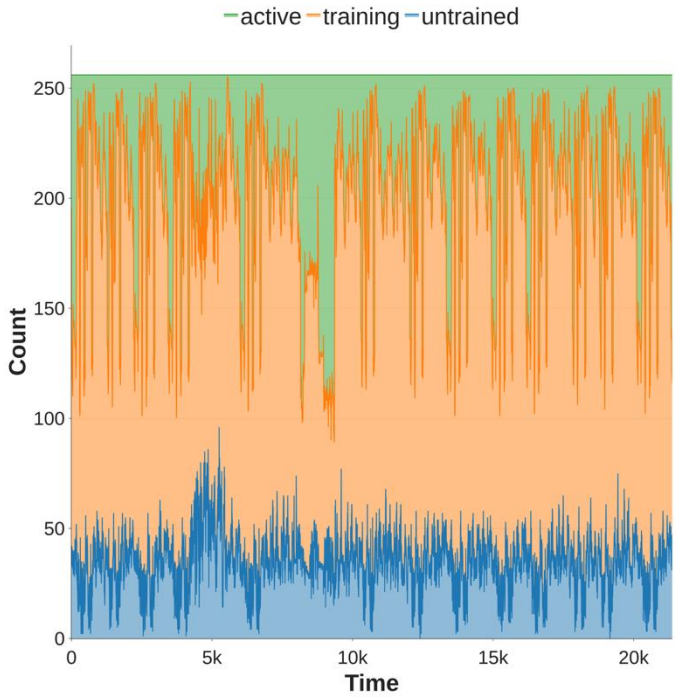


- 2-level value predictor maintains performance at 10% of the size
- The filter is too aggressive for larger table sizes

Evaluation



512-entry prediction table
standard predictor



256-entry filter and 256-entry
main table 2-level predictor

Conclusion

- Value predictors can break true data dependencies
- Need high accuracy, and therefore, only 21% of loads are useful for prediction
- The prediction table uses capacity for many useless loads
- Developed a filter to improve the prediction table behavior
- A size reduction of 90.4%

Future Work:

- Combine with state-of-the-art value predictors
- Combine with criticality factors to create criticality-based value prediction